Gangmin Yu

Mechanical Engineering · Undergraduate Researcher

Kyonggi University · RVLab

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RESEARCH INTERESTS

Autonomous driving and navigation, vehicle dynamics and control, radar/sensor perception, robot simulation and digital twins, embedded and mechatronic systems.

EDUCATION

Kyonggi University, Suwon, Republic of Korea *Mar. 2021 – Present*

B.S. in Mechanical Engineering

Undergraduate Researcher, RVLab (Advisor: Prof. Minseong Choi)

GPA: 4.27 / 4.5 **Major GPA:** 4.41 / 4.5

English: TOEIC Speaking IH (150)

PUBLICATIONS

Domestic Conferences

이충헌, **유강민**, 박영상, 최민성, 조우성. **"Swerve 구동 휴머노이드를 위한 Holonomic 실내 자율 탐색 알고리즘"** *제어·로봇·시스템학회(ICROS) 학술대회, 2026.*

김도윤, 강민수, **유강민**, 서형태, 박영상, 최민성. **"IsaacSim 기반 휴머노이드 ACT 학습제어를 위한 VR 원격조작 및 Jitter 저감 기법"** *제어·로봇·시스템학회(ICROS) 학술대회, 2026.*

성민준, **유강민**, 김준규, 한승호, 양승훈, 최민성**. "도로 노면 의미론적 분할 기법의 소수 클래스 불균형 해소를 위한 원근 변환 기반 데이터 증강 기법"** *제어·로봇·시스템학회(ICROS) 학술대회, 2026.*

RESEARCH & PROJECTS

ONGOING

4D Radar Ego-Velocity Estimation *RVLab, Kyonggi University*

- Estimating ego-vehicle velocity directly from 4D radar measurements toward robust self-motion estimation for autonomous driving.

Manipulation & Navigation for Humanoids in Domestic Environments *RVLab*

- Applied physics-based digital-twin simulation and MetaQuest 3 teleoperation for a door-opening task in Isaac Sim.
- Implemented autonomous driving and navigation using AMCL localization on a generated 2D LiDAR map.

PAST

3D Simulation Environments & Physics Modeling for Daily-Life Scenes *KIMM (commissioned)*

funded by Korea Institute of Machinery & Materials (KIMM)

- Built physically simulated indoor environments in NVIDIA Isaac Sim (PhysX) for realistic robot interaction.
- Authored rigid-body and collider models from .obj meshes; set per-object mass, inertia, friction, restitution.
- Defined articulated structures (hinge/prismatic joints) for doors and drawers with motion limits and dynamics.

Autonomous Vehicle Build *Autonomous Driving Club*

- Led the overall build of an autonomous vehicle; designed the hardware/embedded system (actuators, sensors, MCU communication).
- Implemented steering control (Pure Pursuit) and speed control (PID).

High-Voltage Shutdown Circuit *Self-Built Car Club · Formula (EV)*

- Designed a safety shutdown circuit for 120 V high voltage, cutting power on BMS/IMD/interlock faults or sudden-acceleration detection.
- Designed and fabricated the PCB (KiCad) and applied it to the vehicle.

Braking System Design *Self-Built Car Club · Formula & Baja*

- Computed braking force from vehicle kinematics and designed the brake disc accordingly.
- Used CAE to optimize the design and verify structural safety.
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AWARDS

Excellence Award — Self-Built Autonomous Vehicle Division (Leader) *Nov. 2025*
University Creative Mobility Competition · Korea Transportation Safety Authority (KOTSA)

Acceleration Award — Baja Student (Team Member) *Nov. 2021*
International Self-Built Car Competition · Yeungnam University

TECHNICAL SKILLS

Simulation / Robotics: NVIDIA Isaac Sim, PhysX, ROS2, AMCL / Navigation
Control / Embedded: Embedded C, PID, Pure Pursuit, MCU & Sensors
Design / CAE: KiCad (Schematic / PCB), AutoCAD, CATIA, Ansys
Domain: Autonomous Driving, Vehicle Dynamics, Radar Perception, Mechatronics